

BHRUGU SETLUR

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EDUCATION

New York University Tandon School of Engineering

New York, NY

B.S. Electrical and Computer Engineering | GPA: 3.74 | Dean's List: all semesters, 2025-2026

Expected May 2028

- Coursework: Computer Architecture & Digital Logic (Verilog), Data Structures & Algorithms, Object-Oriented Programming (C++)
- Planned for Fall 2026: Signals & Systems, Operating Systems, Embedded Systems Design

EXPERIENCE

Generex

San Francisco, CA

Embedded Firmware Engineer (Contract)

Jun 2026 - Present

- Developing STM32G071 firmware for a battery monitor: 4 I2C time-of-flight sensors, reporting over an opto-isolated BACS bus.
- Separated protocol and alarm logic from the drivers into portable C, covered by 8,444 host checks that run without target hardware.
- Designed debounce, hysteresis, and confidence-over-time logic to reject false alarms from condensation, bubbles, and detached sensors.
- Validated the BACS CRC across 125,323 frames captured from Generex's official Reader and Simulator, with 0 mismatches.
- Fit the current firmware, excluding diagnostics, into about 12 KB of the chip's 36 KB of RAM.

Tokonoma.ai

San Francisco, CA

Software Engineering Intern

Jun 2026 - Aug 2026

- Prototyped a faster vector-search path, cutting median query time 1.95x on a 20,000-chunk corpus with no measured quality loss.
- Caught a filtered-search slowdown in review and added a tested exact-search fallback plus per-query monitoring.
- Built a labeled benchmark to tune duplicate detection, scoring 81 configurations against matching and non-matching record pairs.
- Built one issue-tracking service for GitHub, GitLab, Jira, and Linear that imports issues and closes them when fixes merge.
- Reviewed the split of a 2,851-line store module, fixing clock-mismatch timeouts and stalled jobs that falsely reported healthy.
- Fixed 5 security vulnerabilities, including prompt injection in an LLM-based extraction path.
- Built passwordless email sign-in and a protected coming-soon dashboard for the company website.

Tech: Python, Rust, PostgreSQL, pgvector, Kubernetes, Helm, Docker, OpenTelemetry, Quickwit, OAuth 2.1, MCP, RAG

Growcery | growcery.us

New York, NY

Software Engineering Intern

Jan 2026 - May 2026

- Launched a two-sided marketplace and integrated Stripe payments with Apple Pay and automatic platform-fee collection.
- Implemented radius-based seller search with Supabase and Mapbox, plus in-app messaging and seller listing tools.

VIP: High Performance Computing at NYU

New York, NY

Virtual Environment Penetration Tester

Sep 2025 - Present

- Ran an authorized 8-phase penetration test in a GCP lab: scanned with nmap, then cracked MD5 hashes from an open FTP share.
- Used SQL injection to dump the database, path traversal to read an SSH key, and command injection to open a reverse shell.
- Recovered credentials from an exposed .git, escalated to root via a sudo misconfiguration, then reused passwords from FTP to SSH.

PROJECTS

Spherra | Compressed Vector Search in Rust | github.com/bhrugu-setlur/spherra

Aug 2026 - Sep 2026

- Built a Rust vector index that compresses 768-dimensional embeddings 6.3x, from 3,072 bytes to 484 bytes per vector.
- Spread values evenly with a randomized Hadamard transform, stored 4-bit codes per coordinate, and product-quantized the error.
- Reached 0.92 to 0.98 recall@10 against exact search on up to 1M real search-query and synthetic embeddings.
- Raised recall@10 from 0.9095 to 0.9255 with no extra storage by dividing each refined score by its rebuilt vector's length.
- Searched 1M vectors in 72 ms median and 114 ms p99 latency on an M1 Pro, and 10M vectors within 3.86 GB of peak memory.
- Verified the fast scan kernel against a reference scorer on 2,073,760 scores with 0 differences.

Tech: Rust, Product Quantization, k-means, Hadamard Transform, MS MARCO, DPR, SciFact, MovieLens, synthetic correlated vectors

8-bit CPU with Virtual Memory & Cache | github.com/bhrugu-setlur/cpu-design

Jun 2026 - Aug 2026

- Designed an 8-bit Harvard CPU in Verilog with a 12-instruction ISA, a direct-mapped L1 cache, and a 4-way LRU L2 cache.
- Added virtual memory with a 4-entry TLB and hardware page-table walks, tested by 15 Verilog suites and on a Basys 3 FPGA.
- Flushed the TLB on page-table writes and froze the CPU on page faults, showing the faulting address on the Basys 3 LEDs.
- Hardened the core to GDS on SkyWater 130 nm with OpenROAD: 12,778 cells, clean DRC, and project-deck LVS on all 146 pairs.
- Met the 10 MHz timing target with 90.29 ns of worst setup slack, and wrote a two-pass C++ assembler for the CPU's programs.
- **Open source:** Merged a regression test into **cocotb**, a widely used Python framework for RTL verification.

Tech: Verilog, RTL Design, FPGA (Xilinx Artix-7), Vivado, Icarus Verilog, cocotb, GTKWave, OpenROAD, Sky130, C++

SKILLS

Languages: Verilog, C, C++, Python, Rust, TypeScript, SQL, Java, C#

AI/ML: RAG, semantic search, vector embeddings (pgvector), HNSW, LLM agents, Model Context Protocol (MCP)

Frameworks & Tools: React, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, Kubernetes, Helm, Docker, OpenTelemetry, Git, GCP, Vercel, Nmap, sqlmap

Hardware & Verification: RTL Design, ASIC Physical Design (OpenROAD, Sky130), STA, DRC, LVS, FPGA (Xilinx Artix-7), Vivado, Icarus Verilog, cocotb, GTKWave, STM32 (ARM Cortex-M0+), I2C, UART

AWARDS AND HONORS

3rd Place, Code4City Hackathon | 6th Place, USA Fencing Div. II Nationals